Expected Value

Contents

1 Definition of Expectation (Discrete and Continuous)

Definition of the Expected Value

The expected value (or *expectation*) of a random variable is one of the most fundamental concepts in probability theory and statistics. It represents the average or center of the probability distribution and can be interpreted as the value that would be obtained by averaging the outcomes of the random variable over an indefinitely large number of independent repetitions of the experiment.

More formally, the expected value is a weighted average of all possible values of the random variable, where each value is weighted according to its probability of occurring.

The precise definition depends on whether the random variable is discrete or continuous.

Discrete Random Variables Suppose that X is a discrete random variable taking values

$$x_1, x_2, x_3, \ldots$$

with probability mass function

$$P(X = x_i) = p_i, \qquad i = 1, 2, \ldots,$$

where

$$p_i \geq 0, \qquad \sum_{i=1}^{\infty} p_i = 1.$$

If the series

$$\sum_{i=1}^{\infty} |x_i| p_i$$

is finite, then the expected value of X is defined as

$$\boxed{E[X] = \sum_{i=1}^{\infty} x_i p_i.}$$

Equivalently,

$$\boxed{E[X] = \sum_{x} x\, P(X = x),}$$

where the summation extends over all possible values of X.

Observe that each possible value contributes proportionally to its probability. Values with larger probabilities contribute more heavily to the expected value than values with smaller probabilities.

Example Consider the random variable

$$X = \begin{cases} 1, & P = \dfrac{1}{4}, \\ 2, & P = \dfrac{1}{2}, \\ 5, & P = \dfrac{1}{4}. \end{cases}$$

Then

$$
\begin{aligned}
E[X] &= 1\left(\frac{1}{4}\right) + 2\left(\frac{1}{2}\right) + 5\left(\frac{1}{4}\right) \\
&= \frac{1}{4} + 1 + \frac{5}{4} \\
&= \frac{10}{4} \\
&= 2.5.
\end{aligned}
$$

Notice that the expected value need not be one of the possible values of the random variable. Although X can only take the values

$$1,\ 2,\ \text{or } 5,$$

its expected value equals

$$2.5.$$

The expectation should therefore be interpreted as a long-run average rather than as a value that the random variable must actually attain.

Continuous Random Variables Suppose now that X is a continuous random variable having probability density function

$$f_X(x),$$

where

$$f_X(x) \geq 0,$$

and

$$\int_{-\infty}^{\infty} f_X(x)\,dx = 1.$$

If

$$\int_{-\infty}^{\infty} |x| f_X(x)\,dx < \infty,$$

then the expected value of X is defined as

$$\boxed{E[X] = \int_{-\infty}^{\infty} x\, f_X(x)\,dx.}$$

This definition is the continuous analogue of the discrete case.
Instead of summing over all possible values, we integrate over the entire real line.
Again, each possible value of X is weighted by the probability density at that point.

Example Suppose

$$X \sim U(0,1),$$

so that

$$f_X(x) = \begin{cases} 1, & 0 \le x \le 1, \\ 0, & \text{otherwise}. \end{cases}$$

The expected value is

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x\, f_X(x)\, dx \\ &= \int_0^1 x\, dx \\ &= \left[\frac{x^2}{2}\right]_0^1 \\ &= \frac{1}{2}. \end{aligned}$$

Thus,

$$\boxed{E[X] = \frac{1}{2}.}$$

As expected, the center of the uniform distribution over the interval $[0,1]$ is precisely its midpoint.

Existence of the Expected Value The expected value is not always finite.

For example, if

$$P(X=n) = \frac{6}{\pi^2 n^2}, \qquad n = 1, 2, \ldots,$$

then

$$\sum_{n=1}^{\infty} P(X=n) = 1,$$

so this is a valid probability distribution.

However,

$$E[X] = \sum_{n=1}^{\infty} n \cdot \frac{6}{\pi^2 n^2} = \frac{6}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n}.$$

Since the harmonic series diverges,

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty,$$

it follows that

$$\boxed{E[X] = \infty.}$$

For this reason, the definition of expectation always requires the additional condition

$$\sum |x| P(X = x) < \infty$$

or

$$\int |x| f(x)\, dx < \infty,$$

which guarantees that the expectation is well defined.

Unified Definition The discrete and continuous definitions can be viewed as special cases of a single principle.

The expected value is obtained by averaging a quantity with respect to the probability distribution of the random variable.

Symbolically,

$$\boxed{E[X] = \int x\, dP,}$$

where the integral is understood in the sense of probability theory.

This unified viewpoint allows all subsequent properties of expectation to be proved simultaneously for both discrete and continuous random variables.

2 Expectation of a Function (LOTUS)

Expectation of a Function (The Law of the Unconscious Statistician)

In many applications, we are interested not only in the expected value of a random variable X, but also in the expected value of some function of that random variable,

$$g(X),$$

where g is a real-valued function.

For example, we may wish to compute

$$E[X^2], \qquad E[e^X], \qquad E[\ln X], \qquad E[\sin X],$$

or any other transformation of the original random variable.

A natural question therefore arises:

Must we first determine the probability distribution of $g(X)$ before computing its expected value?

Fortunately, the answer is no.

The expected value of a function of a random variable can be computed directly from the probability distribution of the original random variable. This remarkable result is known as the **Law of the Unconscious Statistician (LOTUS).**

Discrete Random Variables Suppose that X is a discrete random variable taking values

$$x_1, x_2, \dots$$

with probability mass function

$$P(X = x_i) = p_i.$$

The random variable

$$g(X)$$

takes the values

$$g(x_1), g(x_2), \dots,$$

possibly with repeated values.

Instead of determining the probability distribution of $g(X)$, we compute its expectation directly.

The expected value is defined by

$$\boxed{E[g(X)] = \sum_x g(x)P(X = x),}$$

provided that

$$\sum_x |g(x)|P(X = x) < \infty.$$

Observe that this formula is identical to the definition of expectation, except that the values x have simply been replaced by $g(x)$.

Example Suppose

$$P(X = 1) = 0.2, \qquad P(X = 2) = 0.5, \qquad P(X = 4) = 0.3.$$

Compute

$$E[X^2].$$

Rather than finding the distribution of X^2, we apply LOTUS directly,

$$
\begin{aligned}
E[X^2] &= 1^2(0.2) + 2^2(0.5) + 4^2(0.3) \\
&= 0.2 + 2 + 4.8 \\
&= 7.
\end{aligned}
$$

Notice that the probability distribution of X^2 never had to be computed.

Continuous Random Variables Suppose now that X is a continuous random variable with probability density function

$$f_X(x).$$

Then the expected value of $g(X)$ is

$$\boxed{E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\,dx,}$$

provided that

$$\int_{-\infty}^{\infty} |g(x)| f_X(x)\,dx < \infty.$$

Again, the distribution of $g(X)$ is not required.

Instead, the function g is evaluated at every possible value of X, and the result is averaged using the density of X.

Example Suppose

$$X \sim U(0,1),$$

whose density is

$$f(x) = \begin{cases} 1, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Compute

$$E[X^2].$$

Using LOTUS,

$$
\begin{aligned}
E[X^2] &= \int_0^1 x^2\,dx \\
&= \left[\frac{x^3}{3}\right]_0^1 \\
&= \frac{1}{3}.
\end{aligned}
$$

Thus,

$$\boxed{E[X^2] = \frac{1}{3}.}$$

Again, there was no need to determine the probability density of the random variable X^2.

Why Does LOTUS Work? (Discrete Case) Suppose that

$$Y = g(X).$$

By definition,

$$E[Y] = \sum_y yP(Y = y).$$

Since

$$Y = g(X),$$

the event

$$Y = y$$

occurs whenever

$$g(X) = y.$$

Hence,

$$P(Y = y) = P(g(X) = y).$$

Substituting,

$$E[Y] = \sum_y yP(g(X) = y).$$

Now collect together all values of X that satisfy

$$g(x) = y.$$

Since

$$P(g(X) = y) = \sum_{x:g(x)=y} P(X = x),$$

we obtain

$$E[Y] = \sum_{y} y \sum_{x:g(x)=y} P(X = x)$$

$$= \sum_{x} g(x)P(X = x).$$

Therefore,

$$\boxed{E[g(X)] = \sum_{x} g(x)P(X = x).}$$

Thus, the probability distribution of $g(X)$ never needs to be computed explicitly.

Why Is It Called the "Law of the Unconscious Statistician"? The name originates from the fact that many people compute

$$E[g(X)] = \sum_{x} g(x)P(X = x)$$

or

$$E[g(X)] = \int g(x)f(x)\,dx$$

without consciously realizing that they are using a theorem.

Instead of first determining the probability distribution of $g(X)$, they simply replace the variable x in the expectation formula by $g(x)$ and proceed with the calculation.

LOTUS guarantees that this procedure is mathematically correct.

Special Cases Several important expectations are immediate consequences of LOTUS.

Choosing

$$g(x) = x,$$

gives

$$E[g(X)] = E[X].$$

Choosing

$$g(x) = x^2,$$

gives

$$E[X^2].$$

Choosing

$$g(x) = |x|,$$

gives

$$E[|X|].$$

Choosing

$$g(x) = e^x,$$

gives

$$E[e^X].$$

Thus, virtually every expectation encountered in probability and statistics is an application of LOTUS.

Conclusion The Law of the Unconscious Statistician provides a powerful computational tool.

Rather than determining the probability distribution of a transformed random variable $g(X)$, one computes its expectation directly from the probability distribution of the original random variable.

For a discrete random variable,

$$\boxed{E[g(X)] = \sum_x g(x)P(X = x),}$$

while for a continuous random variable,

$$\boxed{E[g(X)] = \int_{-\infty}^{\infty} g(x)f_X(x)\,dx.}$$

Nearly every property of expectation developed in the following sections is a direct consequence of these two formulas.

3 Expectation of a Constant

Expectation of a Constant

One of the simplest yet most fundamental properties of the expected value is that the expectation of a constant random variable is the constant itself.

Intuitively, if a random variable always takes the same value, then there is no randomness to average. Consequently, its expected value must equal that constant.

We now prove this result separately for discrete and continuous random variables.

Theorem Let

$$X = c,$$

where c is a fixed real number.

Then

$$\boxed{E[c] = c.}$$

Proof for a Discrete Random Variable Suppose that the random variable always takes the value

$$c.$$

Its probability mass function is therefore

$$P(X = c) = 1,$$

and

$$P(X = x) = 0, \qquad x \neq c.$$

By the definition of expectation,

$$E[X] = \sum_x xP(X = x).$$

Since every probability is zero except at $x = c$,

$$E[X] = cP(X = c) + \sum_{x \neq c} xP(X = x)$$

$$= c(1) + 0$$

$$= c.$$

Hence,

$$\boxed{E[c] = c.}$$

Proof for a Continuous Random Variable Although a constant random variable is technically not continuous (its distribution is concentrated at a single point), it is instructive to see how the same conclusion follows formally.

Suppose that

$$X = c.$$

Its probability distribution places all its probability mass at the point c. In measure-theoretic language,

$$f_X(x) = \delta(x - c),$$

where δ denotes the Dirac delta distribution.

Applying the definition of expectation,

$$E[X] = \int_{-\infty}^{\infty} x\,\delta(x - c)\,dx$$

$$= c.$$

Thus,

$$\boxed{E[c] = c.}$$

For an introductory course, the discrete proof is sufficient. The measure-theoretic formulation above simply confirms that the same property holds in complete generality.

Alternative Proof Using LOTUS The result also follows immediately from the Law of the Unconscious Statistician.

Let

$$g(x) = c,$$

that is, g is the constant function.

Applying LOTUS,

$$E[g(X)] = \sum_x g(x)P(X = x)$$

for a discrete random variable, or

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f_X(x)\,dx$$

for a continuous random variable.

Since

$$g(x) = c$$

for every value of x,

$$
\begin{aligned}
E[g(X)] &= \sum_x cP(X = x) \\
&= c\sum_x P(X = x) \\
&= c(1) \\
&= c.
\end{aligned}
$$

Similarly, in the continuous case,

$$
\begin{aligned}
E[g(X)] &= \int_{-\infty}^{\infty} cf_X(x)\,dx \\
&= c\int_{-\infty}^{\infty} f_X(x)\,dx \\
&= c(1) \\
&= c.
\end{aligned}
$$

Therefore,

$$\boxed{E[c] = c.}$$

Example Suppose a machine always produces an object weighing exactly

$$5 \text{ kg}.$$

The corresponding random variable satisfies

$$X = 5.$$

Its expectation is therefore

$$E[X] = 5.$$

Although expectation is often interpreted as a long-run average, in this case every observation equals 5, so the average is trivially equal to 5.

Importance of the Result Although elementary, this property is fundamental because it serves as one of the building blocks of the linearity of expectation.

Indeed, when proving

$$E[aX + b] = aE[X] + b,$$

the term

$$E[b]$$

appears naturally. The present theorem immediately yields

$$E[b] = b,$$

allowing the linearity formula to be established.

Thus, the expectation of a constant is not merely a simple observation; it is an essential ingredient in many subsequent results concerning expected values.

4 Linearity of Expectation

Linearity of Expectation

The most important property of the expected value is its *linearity*. Unlike many other statistical quantities, such as the variance,

$$\mathrm{Var}(X + Y) \neq \mathrm{Var}(X) + \mathrm{Var}(Y)$$

in general.

Expectation behaves much more simply.

The expectation of a linear combination of random variables is the corresponding linear combination of their expectations.

Remarkably, this property holds whether or not the random variables are independent.

This fact makes expectation one of the most useful operators in probability and statistics.

Theorem Let

$$X \quad \text{and} \quad Y$$

be random variables whose expectations exist.

Let

$$a, b \in \mathbb{R}$$

be constants.

Then

$$\boxed{E[aX + bY] = aE[X] + bE[Y].}$$

More generally, if

$$X_1, \ldots, X_n$$

are random variables with finite expectations, then

$$\boxed{E\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i E[X_i].}$$

Notice that no assumption of independence is required.

Proof for Discrete Random Variables Suppose that

$$X \qquad \text{and} \qquad Y$$

have joint probability mass function

$$P(X = x, Y = y).$$

By the definition of expectation,

$$E[aX + bY] = \sum_x \sum_y (ax + by) P(X = x, Y = y).$$

Using the distributive property,

$$\begin{aligned} E[aX + bY] &= \sum_x \sum_y (ax + by)\, P(X = x, Y = y) \\ &= \sum_x \sum_y axP(X = x, Y = y) \\ &\quad + \sum_x \sum_y byP(X = x, Y = y). \end{aligned}$$

Since a and b are constants,

$$\begin{aligned} E[aX + bY] = a \sum_x \sum_y xP(X = x, Y = y) \\ + b \sum_x \sum_y yP(X = x, Y = y). \end{aligned}$$

Now consider the first double summation.
Using the law of total probability,

$$\sum_y P(X = x, Y = y) = P(X = x).$$

Therefore,

$$\sum_x \sum_y xP(X=x,Y=y) = \sum_x x\left(\sum_y P(X=x,Y=y)\right)$$

$$= \sum_x xP(X=x)$$

$$= E[X].$$

Similarly,

$$\sum_x \sum_y yP(X=x,Y=y) = \sum_y y\left(\sum_x P(X=x,Y=y)\right)$$

$$= \sum_y yP(Y=y)$$

$$= E[Y].$$

Substituting these results into the previous expression,

$$E[aX+bY] = aE[X] + bE[Y].$$

Hence,

$$\boxed{E[aX+bY] = aE[X] + bE[Y].}$$

Proof for Continuous Random Variables Suppose that

$$X \qquad \text{and} \qquad Y$$

have joint probability density function

$$f(x,y).$$

The expectation of

$$aX+bY$$

is

$$E[aX+bY] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(ax+by)f(x,y)\,dx\,dy.$$

Using the linearity of integration,

$$E[aX+bY] = a\iint xf(x,y)\,dx\,dy$$
$$+\, b\iint yf(x,y)\,dx\,dy.$$

Evaluating the First Integral Let us carefully study the first double integral,

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x\, f(x,y)\, dx\, dy.$$

Notice that the integrand is

$$x\, f(x,y),$$

where the variable x depends only on x, while the joint density $f(x,y)$ depends on both variables.

Our objective is to show that this double integral is precisely the expectation of the random variable X.

The first step is to change the order of integration.

Since the joint density is nonnegative and integrable, Fubini's Theorem allows us to interchange the order of integration:

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x\, f(x,y)\, dx\, dy = \int_{-\infty}^{\infty}\left(\int_{-\infty}^{\infty} x\, f(x,y)\, dy\right) dx.$$

Notice that the integration is now performed with respect to y first.

Next, observe that during the inner integration,

$$\int_{-\infty}^{\infty} x\, f(x,y)\, dy,$$

the variable x is treated as a constant because the integration variable is y.

Therefore,

$$\int_{-\infty}^{\infty} x\, f(x,y)\, dy = x \int_{-\infty}^{\infty} f(x,y)\, dy.$$

Substituting this into the previous expression gives

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x\, f(x,y)\, dx\, dy = \int_{-\infty}^{\infty} x\left(\int_{-\infty}^{\infty} f(x,y)\, dy\right) dx.$$

Deriving the Marginal Density The quantity

$$f(x,y)$$

is the *joint probability density function* of the random variables X and Y.

It describes how probability is distributed over the two-dimensional (x,y)-plane.

Suppose we are interested only in the random variable X. In other words, we no longer care about the value of Y.

To obtain the probability density of X alone, we must account for *every possible value* that Y can take.

This is accomplished by integrating the joint density over the entire range of Y:

$$\boxed{f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy.}$$

This function is called the *marginal density* of X.

Why Does This Formula Work? Recall that for continuous random variables,

$$P(a \le X \le b,\, c \le Y \le d) = \int_a^b \int_c^d f(x,y)\,dy\,dx.$$

Now suppose we wish to compute the probability that

$$X$$

lies between a and b, regardless of the value of Y.
Since Y may take any value on the real line, we integrate over all possible values of Y:

$$P(a \le X \le b) = \int_a^b \int_{-\infty}^{\infty} f(x,y)\,dy\,dx.$$

On the other hand, by the definition of the marginal density,

$$P(a \le X \le b) = \int_a^b f_X(x)\,dx.$$

Since these two expressions represent exactly the same probability for every interval $[a,b]$,

$$\int_a^b f_X(x)\,dx = \int_a^b \left(\int_{-\infty}^{\infty} f(x,y)\,dy \right) dx.$$

The Fundamental Theorem of Calculus then implies that the integrands must be equal almost everywhere. Therefore,

$$\boxed{f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy.}$$

Geometric Interpretation The joint density

$$f(x,y)$$

may be visualized as a surface lying above the (x,y)-plane.
For a fixed value of x, the function

$$y \mapsto f(x,y)$$

is a vertical slice through this surface.
Integrating

$$f(x, y)$$

with respect to y computes the area under this slice.
That area is precisely the value of the marginal density

$$f_X(x).$$

Thus,

$$f_X(x)$$

measures the total probability density associated with the value x, after accounting for every possible value of the second variable Y.

Example Consider the joint density

$$f(x, y) = \begin{cases} 2, & 0 < x < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

To obtain the marginal density of X, we integrate over all admissible values of Y.
Since the condition

$$0 < x < y < 1$$

implies that

$$y \in (x, 1),$$

we have

$$\begin{aligned} f_X(x) &= \int_x^1 2\, dy \\ &= 2(1 - x), \qquad 0 < x < 1. \end{aligned}$$

Therefore,

$$\boxed{f_X(x) = \begin{cases} 2(1 - x), & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}}$$

Notice that the variable Y has completely disappeared. All of its possible values have been "summed out" through integration, leaving a density that depends only on X.

Intuitively, this integral "adds up" the probability density corresponding to every possible value of Y, leaving only the probability density of X.

Replacing the inner integral by the marginal density,

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x\, f(x, y)\, dx\, dy = \int_{-\infty}^{\infty} x f_X(x)\, dx.$$

Finally, recall the definition of the expectation of a continuous random variable:

$$E[X] = \int_{-\infty}^{\infty} x f_X(x)\,dx.$$

Therefore,

$$\boxed{\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x\,f(x,y)\,dx\,dy = E[X].}$$

The same argument applied to the second integral yields

$$\boxed{\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} y\,f(x,y)\,dx\,dy = E[Y].}$$

Consequently,

$$E[aX + bY] = aE[X] + bE[Y].$$

Generalization The previous result extends naturally to any finite number of random variables.

Indeed,

$$\begin{aligned} E[a_1X_1 + \cdots + a_nX_n] &= E[a_1X_1] + \cdots + E[a_nX_n] \\ &= a_1E[X_1] + \cdots + a_nE[X_n]. \end{aligned}$$

Hence,

$$\boxed{E\left[\sum_{i=1}^{n} a_iX_i\right] = \sum_{i=1}^{n} a_iE[X_i].}$$

Special Cases Several useful identities are immediate consequences of linearity.

Choosing

$$a = 1, \qquad b = 1,$$

gives

$$\boxed{E[X + Y] = E[X] + E[Y].}$$

Choosing

$$a = 1, \qquad b = -1,$$

gives

$$\boxed{E[X - Y] = E[X] - E[Y].}$$

Choosing

$$b = 0,$$

gives

$$\boxed{E[aX] = aE[X].}$$

Choosing

$$a = 1,$$

and letting $Y = c$ be a constant,

$$\begin{aligned} E[X + c] &= E[X] + E[c] \\ &= E[X] + c. \end{aligned}$$

Thus,

$$\boxed{E[X + c] = E[X] + c.}$$

Example Suppose

$$E[X] = 5, \qquad E[Y] = 2.$$

Compute

$$E[3X - 4Y + 7].$$

Applying linearity,

$$\begin{aligned} E[3X - 4Y + 7] &= 3E[X] - 4E[Y] + E[7] \\ &= 3(5) - 4(2) + 7 \\ &= 15 - 8 + 7 \\ &= 14. \end{aligned}$$

$$\boxed{E[3X - 4Y + 7] = 14.}$$

Importance of Linearity Linearity of expectation is one of the most powerful tools in probability theory because it holds without requiring independence among the random variables.

This property greatly simplifies the computation of expectations and plays a central role in statistics.

For example, it allows us to show that the sample mean,

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i,$$

satisfies

$$E[\bar{X}] = \frac{1}{n}\sum_{i=1}^{n} E[X_i].$$

If the observations are identically distributed with mean

$$\mu,$$

then

$$E[\bar{X}] = \frac{1}{n}(n\mu) = \mu,$$

showing that the sample mean is an unbiased estimator of the population mean.

5 Expectation of Sums

Expectation of Sums

One of the most useful consequences of the linearity of expectation is that the expectation of the sum of several random variables is equal to the sum of their expectations.

This property is extremely important in probability and statistics because it allows us to compute the expected value of complicated random quantities by studying each component separately.

Unlike many other probabilistic results, *no independence assumptions are required.* The result is true for every collection of random variables whose expectations exist.

Theorem Let

$$X_1, X_2, \ldots, X_n$$

be random variables with finite expectations.

Then

$$\boxed{E[X_1 + X_2 + \cdots + X_n] = E[X_1] + E[X_2] + \cdots + E[X_n].}$$

Equivalently,

$$\boxed{E\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} E[X_i].}$$

Proof The result follows directly from the linearity of expectation.

Beginning with

$$X_1 + X_2 + \cdots + X_n,$$

we apply the linearity property repeatedly.

For two random variables,

$$E[X_1 + X_2] = E[X_1] + E[X_2].$$

Applying linearity once more,

$$\begin{aligned} E[X_1 + X_2 + X_3] &= E[(X_1 + X_2) + X_3] \\ &= E[X_1 + X_2] + E[X_3] \\ &= E[X_1] + E[X_2] + E[X_3]. \end{aligned}$$

Continuing this process,

$$\begin{aligned} E[X_1 + \cdots + X_n] &= E[(X_1 + \cdots + X_{n-1}) + X_n] \\ &= E[X_1 + \cdots + X_{n-1}] + E[X_n] \\ &= E[X_1] + \cdots + E[X_n]. \end{aligned}$$

Hence,

$$\boxed{E\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} E[X_i].}$$

An Alternative Proof The result also follows immediately from the general linearity theorem.

Indeed,

$$\sum_{i=1}^{n} X_i = 1X_1 + 1X_2 + \cdots + 1X_n.$$

Applying

$$E\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i E[X_i],$$

with

$$a_i = 1 \qquad (i = 1, \ldots, n),$$

gives

$$E\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} E[X_i].$$

No Independence Is Required A common misconception is that the random variables must be independent for linearity of expectation to hold.

This is **not true**.

The identity

$$\boxed{E[X_1 + \cdots + X_n] = E[X_1] + \cdots + E[X_n]}$$

holds for *every* collection of random variables whose expectations exist, regardless of whether they are independent, correlated, or even deterministically related.

The reason is that expectation is a *linear operator*. Its definition depends only on averaging the values of the random variables and does not involve any interaction terms between them.

To illustrate this, suppose that

$$X_1 = X_2 = \cdots = X_n = X.$$

The random variables are perfectly positively correlated; knowing one of them determines all the others exactly.

Then,

$$X_1 + \cdots + X_n = nX.$$

Applying the linearity of expectation,

$$\begin{aligned} E[X_1 + \cdots + X_n] &= E[nX] \\ &= nE[X]. \end{aligned}$$

On the other hand,

$$\begin{aligned} E[X_1] + \cdots + E[X_n] &= E[X] + \cdots + E[X] \\ &= nE[X]. \end{aligned}$$

Hence,

$$\boxed{E[X_1 + \cdots + X_n] = E[X_1] + \cdots + E[X_n],}$$

even though the variables are as dependent as possible.

What Changes When Variables Are Dependent? Although dependence does not affect the expectation of a sum, it does affect its variance.

Recall that

$$\mathrm{Var}(X) = E[(X - E[X])^2].$$

For two random variables,

$$\boxed{\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\ \mathrm{Cov}(X,Y),}$$

where

$$\mathrm{Cov}(X,Y) = E[(X - E[X])(Y - E[Y])]$$

is the covariance between X and Y.

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Proof of the Variance of a Sum We wish to prove that

$$\boxed{\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\ \mathrm{Cov}(X,Y).}$$

This identity is valid for *all* random variables having finite second moments. No independence assumption is required.

Step 1: Recall the Definition of Variance By definition,

$$\mathrm{Var}(Z) = E\left[(Z - E[Z])^2\right].$$

Applying this definition to the random variable

$$Z = X + Y,$$

we obtain

$$\mathrm{Var}(X+Y) = E\left[((X+Y) - E[X+Y])^2\right].$$

Step 2: Use Linearity of Expectation Since expectation is linear,

$$E[X+Y] = E[X] + E[Y].$$

Substituting,

$$\mathrm{Var}(X+Y) = E\left[(X + Y - E[X] - E[Y])^2\right].$$

Grouping terms,

$$\mathrm{Var}(X+Y) = E\Big[((X - E[X]) + (Y - E[Y]))^2\Big].$$

Step 3: Expand the Square Recall the algebraic identity

$$(a+b)^2 = a^2 + 2ab + b^2.$$

Let

$$a = X - E[X], \qquad b = Y - E[Y].$$

Then

$$\mathrm{Var}(X+Y) = E\big[(X - E[X])^2 + 2(X - E[X])(Y - E[Y]) + (Y - E[Y])^2\big].$$

Step 4: Apply Linearity of Expectation Using the linearity of expectation,

$$\begin{aligned}\mathrm{Var}(X+Y) = {} & E[(X - E[X])^2] \\ & + 2E[(X - E[X])(Y - E[Y])] \\ & + E[(Y - E[Y])^2].\end{aligned}$$

Step 5: Recognize Each Term By definition,

$$E[(X - E[X])^2] = \mathrm{Var}(X),$$

and

$$E[(Y - E[Y])^2] = \mathrm{Var}(Y).$$

Moreover, the covariance is defined as

$$\boxed{\mathrm{Cov}(X,Y) = E[(X - E[X])(Y - E[Y])].}$$

Therefore,

$$E[(X - E[X])(Y - E[Y])] = \mathrm{Cov}(X,Y).$$

Substituting these identities,

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + 2\,\mathrm{Cov}(X,Y) + \mathrm{Var}(Y).$$

Rearranging the terms,

$$\boxed{\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X,Y).}$$

This completes the proof.

Special Case: Independent Random Variables If X and Y are independent, then

$$\operatorname{Cov}(X,Y) = 0.$$

Consequently,

$$\boxed{\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y).}$$

This is the familiar variance formula for independent random variables.

Remark This proof highlights an important contrast between expectation and variance.

Expectation is always linear:

$$E[X+Y] = E[X] + E[Y],$$

regardless of any dependence between the variables.

Variance, however, is generally *not* linear because of the covariance term, $2\operatorname{Cov}(X,Y)$, which measures the extent to which the two random variables vary together. Only when this covariance is zero (for example, under independence) does the variance of a sum reduce to the sum of the variances.

More generally, for n random variables,

$$\boxed{\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \operatorname{Var}(X_i) + 2\sum_{i<j} \operatorname{Cov}(X_i, X_j).}$$

Unlike expectation, the variance depends explicitly on the covariance between the variables.

The Independent Case If the random variables are independent, then

$$\operatorname{Cov}(X_i, X_j) = 0, \qquad i \neq j.$$

Consequently,

$$\boxed{\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \operatorname{Var}(X_i).}$$

This simplification is one of the main reasons why independence is such an important assumption in probability and statistics.

The Dependent Case When the variables are dependent, the covariance terms generally do not vanish.

For example, suppose

$$X_1 = X_2 = \cdots = X_n = X.$$

Then

$$\sum_{i=1}^{n} X_i = nX,$$

so

$$\begin{aligned}\operatorname{Var}(X_1 + \cdots + X_n) &= \operatorname{Var}(nX) \\ &= n^2 \operatorname{Var}(X).\end{aligned}$$

However,

$$\sum_{i=1}^{n} \operatorname{Var}(X_i) = n \operatorname{Var}(X),$$

which is much smaller whenever $n > 1$.

The difference is explained by the covariance terms.

Indeed,

$$\operatorname{Cov}(X_i, X_j) = \operatorname{Var}(X), \qquad i \neq j,$$

so

$$\begin{aligned}\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) &= n \operatorname{Var}(X) + 2\binom{n}{2} \operatorname{Var}(X) \\ &= n \operatorname{Var}(X) + n(n-1) \operatorname{Var}(X) \\ &= n^2 \operatorname{Var}(X),\end{aligned}$$

which agrees with the direct calculation.

Conclusion The expectation operator is always linear:

$$\boxed{E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i],}$$

regardless of the dependence among the random variables.

In contrast, the variance of a sum depends on the covariance structure:

$$\boxed{\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \operatorname{Var}(X_i) + 2\sum_{i<j} \operatorname{Cov}(X_i, X_j).}$$

Thus, independence is **not** required for linearity of expectation, but it is essential for the simple variance formula

$$\boxed{\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \operatorname{Var}(X_i).}$$

This distinction is one of the most important conceptual differences between expectation and variance.

Example 1 Suppose

$$E[X_1] = 3, \qquad E[X_2] = -1, \qquad E[X_3] = 5.$$

Then

$$\begin{aligned} E[X_1 + X_2 + X_3] &= 3 + (-1) + 5 \\ &= 7. \end{aligned}$$

Therefore,

$$\boxed{E[X_1 + X_2 + X_3] = 7.}$$

Example 2 Suppose

$$E[X_i] = 10, \qquad i = 1, \dots, 100.$$

Then

$$\begin{aligned} E[X_1 + \cdots + X_{100}] &= 100 \times 10 \\ &= 1000. \end{aligned}$$

Notice that we never needed to know whether the variables were independent.

Application to the Sample Mean One of the most important applications of this theorem is the computation of the expected value of the sample mean.

Recall that

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

Applying the linearity of expectation,

$$\begin{aligned} E[\bar{X}] &= E\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) \\ &= \frac{1}{n}E\left(\sum_{i=1}^{n} X_i\right) \\ &= \frac{1}{n}\sum_{i=1}^{n} E[X_i]. \end{aligned}$$

If the observations are identically distributed with common mean

$$E[X_i] = \mu,$$

then

$$\begin{aligned} E[\bar{X}] &= \frac{1}{n}(\mu + \mu + \cdots + \mu) \\ &= \frac{1}{n}(n\mu) \\ &= \mu. \end{aligned}$$

Therefore,

$$\boxed{E[\bar{X}] = \mu.}$$

This shows that the sample mean is an unbiased estimator of the population mean.

Importance of the Result The expectation of sums is one of the most frequently used tools in probability and statistics.

It appears in the derivation of the sample mean, estimation theory, the Law of Large Numbers, the Central Limit Theorem, regression analysis, stochastic processes, and many other areas.

Its simplicity is one of the reasons why expectation is often much easier to analyze than other quantities such as the variance, whose behavior under addition requires additional assumptions about covariance and independence.

6 Expectation of Products of Independent Variables

Expectation of Products of Independent Random Variables

Unlike the expectation of a sum, the expectation of a product generally cannot be separated into the product of the expectations.

In fact,

$$E[XY] \neq E[X]E[Y]$$

in general.

However, if the random variables are independent, then a remarkable simplification occurs.

Theorem Let

$$X \quad \text{and} \quad Y$$

be independent random variables whose expectations exist.

Then

$$\boxed{E[XY] = E[X]E[Y].}$$

More generally, if

$$X_1, X_2, \ldots, X_n$$

are mutually independent random variables with finite expectations, then

$$\boxed{E\left(\prod_{i=1}^{n} X_i\right) = \prod_{i=1}^{n} E[X_i].}$$

Proof for Discrete Random Variables Suppose that

$$X$$

and

$$Y$$

are discrete random variables.

By the definition of expectation,

$$E[XY] = \sum_x \sum_y xyP(X = x, Y = y).$$

Since X and Y are independent,

$$P(X = x, Y = y) = P(X = x)P(Y = y).$$

Substituting this into the expectation,

$$E[XY] = \sum_x \sum_y xyP(X = x)P(Y = y).$$

Rearranging the terms,

$$E[XY] = \sum_x xP(X = x) \sum_y yP(Y = y).$$

The first summation is simply

$$E[X],$$

while the second summation is

$$E[Y].$$

Therefore,

$$\boxed{E[XY] = E[X]E[Y].}$$

Proof for Continuous Random Variables Suppose now that

$$X$$

and

$$Y$$

are continuous random variables.
The expectation of their product is

$$E[XY] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xyf_{X,Y}(x, y)\, dx\, dy.$$

Since X and Y are independent,

$$f_{X,Y}(x, y) = f_X(x)f_Y(y).$$

Substituting,

$$E[XY] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xyf_X(x)f_Y(y)\, dx\, dy.$$

Since the integrand factors into a function of x multiplied by a function of y, the double integral separates into the product of two single integrals,

$$E[XY] = \left(\int_{-\infty}^{\infty} xf_X(x)\, dx \right) \left(\int_{-\infty}^{\infty} yf_Y(y)\, dy \right).$$

Recognizing these integrals,

$$\boxed{E[XY] = E[X]E[Y].}$$

Generalization to Several Random Variables Suppose

$$X_1, X_2, \ldots, X_n$$

are mutually independent.

Then

$$E[X_1X_2\cdots X_n] = \sum\cdots\sum x_1x_2\cdots x_n P(X_1 = x_1, \ldots, X_n = x_n).$$

Independence implies

$$P(X_1 = x_1, \ldots, X_n = x_n) = \prod_{i=1}^{n} P(X_i = x_i).$$

Hence,

$$E[X_1X_2\cdots X_n] = \sum\cdots\sum \left(x_1P(X_1 = x_1)\right)\cdots\left(x_nP(X_n = x_n)\right).$$

The multiple summation separates into the product of the individual summations,

$$\begin{aligned} E[X_1X_2\cdots X_n] &= \left(\sum x_1P(X_1 = x_1)\right)\cdots\left(\sum x_nP(X_n = x_n)\right) \\ &= E[X_1]E[X_2]\cdots E[X_n]. \end{aligned}$$

Therefore,

$$\boxed{E\left(\prod_{i=1}^{n} X_i\right) = \prod_{i=1}^{n} E[X_i].}$$

Why Independence Is Necessary The previous theorem depends crucially on the independence of the random variables.

If the variables are dependent, the identity generally fails.

For example, let

$$Y = X.$$

Then

$$XY = X^2,$$

so that

$$E[XY] = E[X^2].$$

On the other hand,

$$E[X]E[Y] = (E[X])^2.$$

These two quantities are generally different because

$$E[X^2] = (E[X])^2 + \mathrm{Var}(X).$$

Thus,

$$E[X^2] = (E[X])^2$$

only when

$$\mathrm{Var}(X) = 0,$$

that is, when X is a constant random variable.
This example illustrates that independence is essential.

Example Suppose

$$E[X] = 4, \qquad E[Y] = 3,$$

and assume that X and Y are independent.
Then

$$\begin{aligned} E[XY] &= E[X]E[Y] \\ &= 4 \times 3 \\ &= 12. \end{aligned}$$

Hence,

$$\boxed{E[XY] = 12.}$$

Application to Variance One of the most important applications of this theorem occurs in the study of variance.
Recall that

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2.$$

If X and Y are independent,

$$E[XY] = E[X]E[Y].$$

Consequently,

$$\begin{aligned}\mathrm{Var}(X+Y) &= E[(X+Y)^2] - (E[X+Y])^2 \\ &= E[X^2] + 2E[XY] + E[Y^2] \\ &\quad - (E[X]+E[Y])^2.\end{aligned}$$

Using

$$E[XY] = E[X]E[Y],$$

the cross terms cancel,

$$\boxed{\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).}$$

This important identity would generally be false without the independence assumption.

Summary For sums,

$$\boxed{E[X+Y] = E[X] + E[Y],}$$

and this holds for all random variables.

For products,

$$\boxed{E[XY] = E[X]E[Y],}$$

the identity holds only when the random variables are independent.

This distinction is one of the most important facts in probability theory and should always be kept in mind when manipulating expected values.

7 Monotonicity

Monotonicity of Expectation

One of the most natural properties of the expected value is that it preserves order.

Intuitively, if one random variable is always less than or equal to another, then its average value should also be less than or equal to the average of the larger variable.

This property is known as the *monotonicity of expectation.*

Theorem (Monotonicity) Let

$$X \quad \text{and} \quad Y$$

be random variables such that

$$X \leq Y$$

with probability one.

Then

$$\boxed{E[X] \leq E[Y].}$$

Equivalently,

$$X \leq Y \quad \Longrightarrow \quad E[X] \leq E[Y].$$

Interpretation The condition

$$X \leq Y$$

means that for every possible outcome,

$$X(\omega) \leq Y(\omega).$$

That is, no realization of X ever exceeds the corresponding realization of Y.

Since expectation is an average over all possible realizations, it is natural that the average value of X cannot exceed the average value of Y.

Proof Using Linearity Define the random variable

$$Z = Y - X.$$

Since

$$X \leq Y,$$

we have

$$Z \geq 0$$

for every outcome.

Applying the linearity of expectation,

$$\begin{aligned} E[Z] &= E[Y - X] \\ &= E[Y] - E[X]. \end{aligned}$$

Since Z is always nonnegative,

$$E[Z] \geq 0.$$

Therefore,

$$E[Y] - E[X] \geq 0,$$

which implies

$$\boxed{E[X] \leq E[Y].}$$

This completes the proof.

Direct Proof for Discrete Random Variables Suppose that

$$X$$

has probability mass function

$$P(X = x_i) = p_i,$$

and that

$$g(X) = Y,$$

where

$$g(x) \geq x$$

for every possible value of x.
Then

$$\begin{aligned} E[Y] - E[X] &= \sum_i g(x_i)p_i - \sum_i x_i p_i \\ &= \sum_i (g(x_i) - x_i)p_i. \end{aligned}$$

Since

$$g(x_i) - x_i \geq 0,$$

and

$$p_i \geq 0,$$

every term in the summation is nonnegative.
Hence,

$$E[Y] - E[X] \geq 0,$$

or equivalently,

$$\boxed{E[X] \leq E[Y].}$$

Direct Proof for Continuous Random Variables Suppose

$$X \le Y$$

almost surely.
Then

$$Y - X \ge 0.$$

Therefore,

$$\begin{aligned} E[Y] - E[X] &= E[Y - X] \\ &= \int_{-\infty}^{\infty} (Y - X) f(x)\, dx. \end{aligned}$$

Since

$$Y - X \ge 0$$

and

$$f(x) \ge 0,$$

the integrand is everywhere nonnegative.
Consequently,

$$E[Y] - E[X] \ge 0,$$

which again yields

$$\boxed{E[X] \le E[Y].}$$

Example 1 Suppose

$$X = \begin{cases} 1, & P = \frac{1}{2}, \\ 3, & P = \frac{1}{2}, \end{cases}$$

and define

$$Y = X + 2.$$

Clearly,

$$X \le Y$$

for every outcome.
Now compute the expectations.
First,

$$E[X] = 1\left(\frac{1}{2}\right) + 3\left(\frac{1}{2}\right)$$

$$= 2.$$

Next,

$$E[Y] = E[X+2]$$

$$= E[X] + 2$$

$$= 4.$$

Thus,

$$\boxed{E[X] = 2 < 4 = E[Y].}$$

Example 2 Suppose

$$0 \le X \le 10.$$

Applying monotonicity twice,

$$E[0] \le E[X] \le E[10].$$

Since

$$E[0] = 0, \qquad E[10] = 10,$$

we conclude

$$\boxed{0 \le E[X] \le 10.}$$

Thus, whenever a random variable is bounded, its expectation must lie between the same bounds.

Important Consequences Monotonicity immediately implies several useful inequalities.

If

$$X \ge 0,$$

then

$$0 \le X.$$

Applying monotonicity,

$$\boxed{E[X] \ge 0.}$$

Similarly, if

$$a \leq X \leq b,$$

then

$$\boxed{a \leq E[X] \leq b.}$$

These results are frequently used when proving bounds in probability and statistics.

Importance of Monotonicity Monotonicity is one of the fundamental properties of expectation.

Together with linearity, it forms the basis of many important inequalities, including

- Markov's Inequality,
- Chebyshev's Inequality,
- Jensen's Inequality,
- Hölder's Inequality,
- Minkowski's Inequality.

Many advanced results in probability theory ultimately rely on these two simple properties of expectation.

8 Non-negativity

Non-negativity of Expectation

An immediate consequence of the monotonicity property is that the expected value of a nonnegative random variable is itself nonnegative.

This result is one of the most frequently used properties of expectation, particularly in proving inequalities and establishing bounds in probability theory.

Theorem (Non-negativity) Let

$$X$$

be a random variable satisfying

$$X \geq 0$$

almost surely.

Then

$$\boxed{E[X] \geq 0.}$$

Interpretation The condition

$$X \geq 0$$

means that every possible realization of the random variable is nonnegative.

Since expectation is simply a weighted average of all possible values, an average of nonnegative numbers can never be negative.

Therefore,

$$E[X] \geq 0.$$

Proof Using Monotonicity Since

$$0 \leq X,$$

the monotonicity of expectation immediately gives

$$E[0] \leq E[X].$$

From the property of the expectation of a constant,

$$E[0] = 0.$$

Hence,

$$\boxed{E[X] \geq 0.}$$

This proof is valid for every random variable whose expectation exists.

Direct Proof for Discrete Random Variables Suppose that

$$X$$

is discrete with probability mass function

$$P(X = x_i) = p_i.$$

Since

$$X \geq 0,$$

every possible value satisfies

$$x_i \geq 0.$$

Moreover,

$$p_i \geq 0$$

for every i.
The expectation is

$$E[X] = \sum_i x_i p_i.$$

Each term

$$x_i p_i$$

is the product of two nonnegative numbers.
Therefore,

$$x_i p_i \geq 0$$

for every i.
Since a sum of nonnegative numbers is nonnegative,

$$\boxed{E[X] \geq 0.}$$

Direct Proof for Continuous Random Variables Suppose that

$$X$$

has probability density function

$$f_X(x).$$

The expectation is

$$E[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx.$$

Since

$$X \geq 0$$

almost surely,

$$f_X(x) = 0 \qquad \text{whenever } x < 0.$$

Therefore,

$$E[X] = \int_{0}^{\infty} x f_X(x)\, dx.$$

Both factors satisfy

$$x \geq 0, \qquad f_X(x) \geq 0.$$

Hence,

$$x f_X(x) \geq 0$$

for every x.
The integral of a nonnegative function is itself nonnegative.
Consequently,

$$\boxed{E[X] \geq 0.}$$

Example 1 Suppose

$$X = \begin{cases} 0, & P = \frac{1}{4}, \\ 2, & P = \frac{1}{2}, \\ 6, & P = \frac{1}{4}. \end{cases}$$

Then

$$E[X] = 0\left(\frac{1}{4}\right) + 2\left(\frac{1}{2}\right) + 6\left(\frac{1}{4}\right)$$

$$= 0 + 1 + 1.5$$

$$= 2.5.$$

As predicted,

$$\boxed{E[X] = 2.5 \geq 0.}$$

Example 2 Suppose

$$X \sim U(0,3).$$

Its density is

$$f_X(x) = \frac{1}{3}, \qquad 0 \leq x \leq 3.$$

The expectation is

$$E[X] = \int_0^3 x \left(\frac{1}{3}\right) dx$$
$$= \frac{1}{3} \left[\frac{x^2}{2}\right]_0^3$$
$$= \frac{1}{3} \cdot \frac{9}{2}$$
$$= \frac{3}{2}.$$

Again,

$$\boxed{E[X] = \frac{3}{2} > 0.}$$

Important Consequences The non-negativity property immediately implies several useful results.

If

$$X^2 \geq 0,$$

then

$$\boxed{E[X^2] \geq 0.}$$

Likewise,

$$|X| \geq 0$$

implies

$$\boxed{E[|X|] \geq 0.}$$

Similarly,

$$(X - \mu)^2 \geq 0,$$

where

$$\mu = E[X],$$

gives

$$E[(X - \mu)^2] \geq 0.$$

Since

$$\mathrm{Var}(X) = E[(X-\mu)^2],$$

we immediately conclude

$$\boxed{\mathrm{Var}(X) \geq 0.}$$

Thus, one of the most important properties of variance follows directly from the non-negativity of expectation.

Importance of the Result Although simple, the non-negativity property is fundamental throughout probability and statistics.

It is the starting point for many important inequalities, including

$$\mathrm{Var}(X) \geq 0,$$

Markov's Inequality,
Chebyshev's Inequality,
Jensen's Inequality,
and numerous other results involving expectations of nonnegative random variables.

Together with linearity and monotonicity, it forms one of the basic axioms governing the behavior of the expectation operator.

9 Absolute Value Inequality

Absolute Value Inequality

One of the most useful inequalities involving expected values states that the absolute value of the expectation can never exceed the expectation of the absolute value.

This inequality plays a central role in probability theory, mathematical statistics, and analysis. It is frequently used to bound expectations whose exact values are difficult to compute.

Theorem (Absolute Value Inequality) Let X be a random variable such that

$$E[|X|] < \infty.$$

Then

$$\boxed{|E[X]| \leq E[|X|].}$$

This inequality is sometimes called the *triangle inequality for expectation.*

Interpretation The quantity

$$E[X]$$

is the average value of the random variable.

Positive and negative values of X may cancel one another, causing the expectation to be relatively small.

On the other hand,

$$E[|X|]$$

measures the average magnitude of the random variable.

Since taking absolute values removes all cancellations, it is natural that

$$E[|X|]$$

is always at least as large as

$$|E[X]|.$$

Proof For every real number x,

$$-|x| \leq x \leq |x|.$$

Replacing x by the random variable X,

$$-|X| \leq X \leq |X|.$$

Applying the monotonicity of expectation,

$$E[-|X|] \leq E[X] \leq E[|X|].$$

Using the linearity of expectation,

$$E[-|X|] = -E[|X|].$$

Therefore,

$$-E[|X|] \leq E[X] \leq E[|X|].$$

Thus, the expectation lies inside the interval

$$[-E[|X|], E[|X|]].$$

A basic property of real numbers states that

$$-a \leq b \leq a \qquad (a \geq 0)$$

if and only if

$$|b| \le a.$$

Applying this property with

$$a = E[|X|]$$

and

$$b = E[X],$$

we conclude

$$\boxed{|E[X]| \le E[|X|].}$$

This completes the proof.

Alternative Proof (Discrete Case) Suppose that X is a discrete random variable. Its expectation is

$$E[X] = \sum_x xP(X = x).$$

Taking absolute values,

$$|E[X]| = \left| \sum_x xP(X = x) \right|.$$

Applying the triangle inequality for finite or infinite sums,

$$\left| \sum_x xP(X = x) \right| \le \sum_x |x|P(X = x).$$

Recognizing the last summation,

$$\boxed{|E[X]| \le E[|X|].}$$

Alternative Proof (Continuous Case) Suppose that X has probability density function

$$f_X(x).$$

Then

$$E[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx.$$

Taking absolute values,

$$|E[X]| = \left| \int_{-\infty}^{\infty} x f_X(x)\, dx \right|.$$

Using the triangle inequality for integrals,

$$\left| \int g(x)\, dx \right| \le \int |g(x)|\, dx,$$

with

$$g(x) = x f_X(x),$$

we obtain

$$|E[X]| \le \int_{-\infty}^{\infty} |x| f_X(x)\, dx$$

$$= E[|X|].$$

Therefore,

$$\boxed{|E[X]| \le E[|X|].}$$

Example 1 Suppose

$$X = \begin{cases} -2, & P = \frac{1}{2}, \\ 4, & P = \frac{1}{2}. \end{cases}$$

First,

$$E[X] = (-2)\left(\frac{1}{2}\right) + 4\left(\frac{1}{2}\right)$$

$$= -1 + 2$$

$$= 1.$$

Hence,

$$|E[X]| = 1.$$

Now compute

$$E[|X|].$$

Since

$$|X| = \begin{cases} 2, & P = \frac{1}{2}, \\ 4, & P = \frac{1}{2}, \end{cases}$$

we have

$$\begin{aligned} E[|X|] &= 2\left(\frac{1}{2}\right) + 4\left(\frac{1}{2}\right) \\ &= 1 + 2 \\ &= 3. \end{aligned}$$

Therefore,

$$|E[X]| = 1 < 3 = E[|X|].$$

Example 2 Suppose

$$X = \begin{cases} -1, & P = \frac{1}{2}, \\ 1, & P = \frac{1}{2}. \end{cases}$$

Then

$$E[X] = 0,$$

while

$$E[|X|] = 1.$$

Thus,

$$\boxed{0 = |E[X]| < E[|X|] = 1.}$$

This example illustrates how positive and negative values may cancel in the expectation but not in the expectation of the absolute value.

When Does Equality Hold? Equality holds if and only if the random variable never changes sign.

More precisely,

$$|E[X]| = E[|X|]$$

if and only if either

$$X \geq 0 \quad \text{almost surely,}$$

or

$$X \leq 0 \quad \text{almost surely.}$$

Indeed, if X is always nonnegative,

$$|X| = X,$$

so

$$|E[X]| = E[X] = E[|X|].$$

Similarly, if X is always nonpositive,

$$|X| = -X,$$

and

$$|E[X]| = -E[X] = E[|X|].$$

Whenever the random variable takes both positive and negative values with positive probability, strict inequality generally holds because of cancellation.

Importance of the Result The Absolute Value Inequality is one of the fundamental inequalities of probability theory.

It is frequently used to establish upper bounds on expectations and serves as the foundation for many deeper inequalities in analysis and probability, including Hölder's Inequality, Minkowski's Inequality, Jensen's Inequality, and various convergence theorems.

Together with linearity, monotonicity, and non-negativity, it forms one of the essential algebraic properties of the expectation operator.

10 Indicator Random Variables

Indicator Random Variables

Indicator random variables are among the most useful tools in probability and statistics. They transform logical statements or events into random variables that take only the values 0 and 1.

Their simplicity makes them extremely powerful, especially when computing expectations, probabilities, and counting random quantities.

Many elegant probabilistic proofs rely almost entirely on indicator random variables and the linearity of expectation.

Definition Let

$$A$$

be an event in a probability space.

The *indicator random variable* associated with the event A is defined by

$$\boxed{I_A = \begin{cases} 1, & \text{if } A \text{ occurs,} \\ 0, & \text{if } A \text{ does not occur.} \end{cases}}$$

Thus, an indicator variable simply records whether an event has occurred.

If the event happens,

$$I_A = 1,$$

otherwise,

$$I_A = 0.$$

Probability Distribution Since the indicator variable takes only two possible values,

$$0 \quad \text{and} \quad 1,$$

its probability distribution is particularly simple.

Indeed,

$$P(I_A = 1) = P(A),$$

and

$$P(I_A = 0) = 1 - P(A).$$

Therefore,

$$I_A \sim \text{Bernoulli}(P(A)).$$

Expectation of an Indicator Variable The expectation follows directly from the definition.

Using the discrete expectation formula,

$$E[I_A] = \sum_x x P(I_A = x).$$

Since the indicator takes only the values 0 and 1,

$$\begin{aligned} E[I_A] &= 0 \cdot P(I_A = 0) + 1 \cdot P(I_A = 1) \\ &= 0 \cdot (1 - P(A)) + 1 \cdot P(A) \\ &= P(A). \end{aligned}$$

Hence,

$$\boxed{E[I_A] = P(A).}$$

This is one of the most important identities in probability theory.
It allows probabilities to be computed using expectations.

Variance of an Indicator Variable Since

$$I_A^2 = I_A,$$

we have

$$E[I_A^2] = E[I_A] = P(A).$$

Using the variance formula,

$$\text{Var}(X) = E[X^2] - (E[X])^2,$$

we obtain

$$\begin{aligned}\text{Var}(I_A) &= P(A) - P(A)^2 \\ &= P(A)(1 - P(A)).\end{aligned}$$

Thus,

$$\boxed{\text{Var}(I_A) = P(A)(1 - P(A)).}$$

This is precisely the variance of a Bernoulli random variable.

Indicator Variables and Counting Indicator variables become especially useful when counting objects.

Suppose that

$$A_1, A_2, \dots, A_n$$

are events.

Define

$$I_i = I_{A_i}, \qquad i = 1, \dots, n.$$

Now consider

$$N = I_1 + I_2 + \cdots + I_n.$$

Since each indicator contributes either 0 or 1,

$$N$$

counts the number of events that occur.

Expected Number of Occurring Events Applying the linearity of expectation,

$$\begin{aligned} E[N] &= E[I_1 + \cdots + I_n] \\ &= E[I_1] + \cdots + E[I_n]. \end{aligned}$$

Using

$$E[I_i] = P(A_i),$$

we obtain

$$\boxed{E[N] = \sum_{i=1}^{n} P(A_i).}$$

Notice that no independence assumption is required.
This is one of the most useful consequences of the linearity of expectation.

Example 1: Tossing a Coin Suppose a fair coin is tossed once.
Define

$$I = \begin{cases} 1, & \text{if Heads occurs,} \\ 0, & \text{otherwise.} \end{cases}$$

Since

$$P(\text{Heads}) = \frac{1}{2},$$

we have

$$\boxed{E[I] = \frac{1}{2}.}$$

The expected value of the indicator is exactly the probability of obtaining Heads.

Example 2: Number of Heads in Five Tosses Suppose a fair coin is tossed five times. For each toss, define

$$I_i = \begin{cases} 1, & \text{if toss } i \text{ is Heads,} \\ 0, & \text{otherwise.} \end{cases}$$

The total number of Heads is

$$H = I_1 + I_2 + I_3 + I_4 + I_5.$$

Applying linearity,

$$\begin{aligned} E[H] &= E[I_1] + \cdots + E[I_5] \\ &= 5\left(\frac{1}{2}\right) \\ &= 2.5. \end{aligned}$$

Thus,

$$\boxed{E[H] = 2.5.}$$

Notice that we never needed the distribution of H, which is actually binomial.

Example 3: Expected Number of Fixed Points Suppose a permutation of

$$\{1, 2, \ldots, n\}$$

is chosen uniformly at random.
Let

$$I_i = \begin{cases} 1, & \text{if } i \text{ remains in its original position,} \\ 0, & \text{otherwise.} \end{cases}$$

The total number of fixed points is

$$F = \sum_{i=1}^{n} I_i.$$

Since

$$P(i \text{ is fixed}) = \frac{1}{n},$$

we have

$$E[I_i] = \frac{1}{n}.$$

Therefore,

$$E[F] = \sum_{i=1}^{n} E[I_i]$$

$$= n\left(\frac{1}{n}\right)$$

$$= 1.$$

Hence,

$$\boxed{E[F] = 1.}$$

Remarkably, this result holds for every positive integer n.

Importance of Indicator Variables Indicator random variables provide a bridge between probabilities and expectations through the identity

$$\boxed{E[I_A] = P(A).}$$

Combined with the linearity of expectation, this simple identity allows many counting problems to be solved without computing complicated probability distributions.

Indicator variables are fundamental in combinatorics, graph theory, stochastic processes, randomized algorithms, statistical inference, and many other areas of probability.

For this reason, they are considered one of the most powerful tools in elementary probability theory.

11 Sample Mean is Unbiased

The Sample Mean is an Unbiased Estimator

One of the most fundamental results in statistics is that the sample mean is an *unbiased estimator* of the population mean.

This property explains why the sample mean is the natural estimator of the unknown population mean. On average, it neither systematically overestimates nor underestimates the true parameter.

Definition of the Sample Mean Suppose

$$X_1, X_2, \ldots, X_n$$

are independent and identically distributed (i.i.d.) random variables with

$$E[X_i] = \mu, \qquad i = 1, \ldots, n.$$

The sample mean is defined as

$$\boxed{\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.}$$

Its purpose is to estimate the unknown population mean

$$\mu.$$

Definition of an Unbiased Estimator An estimator

$$\hat{\theta}$$

of a parameter

$$\theta$$

is said to be *unbiased* if

$$\boxed{E[\hat{\theta}] = \theta.}$$

In other words, the average value of the estimator over repeated sampling is exactly the parameter being estimated.

Thus, to prove that the sample mean is unbiased, we must show that

$$E[\bar{X}] = \mu.$$

Step 1: Start from the Definition By definition,

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

Taking expectations on both sides,

$$E[\bar{X}] = E\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right).$$

Step 2: Apply the Constant Multiple Rule Since

$$\frac{1}{n}$$

is a constant, it can be taken outside the expectation,

$$E[\bar{X}] = \frac{1}{n}E\left(\sum_{i=1}^{n} X_i\right).$$

Step 3: Apply the Linearity of Expectation Using the linearity of expectation,

$$E\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} E[X_i].$$

Therefore,

$$E[\bar{X}] = \frac{1}{n}\sum_{i=1}^{n} E[X_i].$$

Notice that this step does **not** require the random variables to be independent. Linearity of expectation always holds.

Step 4: Use Identical Distribution Since the observations are identically distributed,

$$E[X_i] = \mu \qquad (i = 1, \ldots, n).$$

Substituting,

$$E[\bar{X}] = \frac{1}{n}(\mu + \mu + \cdots + \mu).$$

There are exactly n copies of μ, so

$$\mu + \mu + \cdots + \mu = n\mu.$$

Hence,

$$E[\bar{X}] = \frac{1}{n}(n\mu)$$

$$= \mu.$$

Therefore,

$$\boxed{E[\bar{X}] = \mu.}$$

This proves that the sample mean is an unbiased estimator of the population mean.

Interpretation Imagine repeatedly drawing random samples of size n from the same population.

Each sample produces a different sample mean,

$$\bar{X}_1, \bar{X}_2, \bar{X}_3, \ldots$$

Some sample means will be larger than the true population mean, while others will be smaller.

However, if we average all these sample means, we obtain exactly

$$\mu.$$

Thus, the estimator has no systematic tendency to overestimate or underestimate the parameter.

Example Suppose a population has mean

$$\mu = 50.$$

We repeatedly draw random samples of size

$$n = 10.$$

Although the sample means may vary,

$$48.7, \quad 52.1, \quad 49.5, \quad 50.9, \quad 47.8, \quad \ldots,$$

their long-run average is

$$50.$$

Therefore,

$$E[\bar{X}] = 50.$$

The estimator fluctuates because of sampling variability, but it is centered at the true population mean.

Why Independence Is Mentioned The proof itself does not require independence.

Only the assumptions

$$E[X_i] = \mu$$

for all i are needed.

Independence becomes important when studying the variance of the sample mean,

$$\mathrm{Var}(\bar{X}),$$

or when deriving its sampling distribution, such as in the Central Limit Theorem.

Thus,

$$\boxed{\text{Unbiasedness requires only identical means,}}$$

whereas

$$\boxed{\text{Variance formulas and sampling distributions require additional assumptions,}}$$

typically independence.

Importance of the Result The unbiasedness of the sample mean is one of the cornerstones of mathematical statistics.

It justifies using

$$\bar{X}$$

as an estimator of the unknown population mean and serves as the foundation for confidence intervals, hypothesis testing, regression analysis, the Central Limit Theorem, and many other statistical procedures.

Together with its variance,

$$\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n},$$

this result explains why the sample mean becomes increasingly accurate as the sample size grows.

12 Summary Table

Summary Table

The table below summarizes the most important properties of the expectation operator presented in this chapter.

Property	**Formula**
Definition (Discrete)	$E[X] = \sum_x x P(X = x)$
Definition (Continuous)	$E[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx$
Expectation of a Function (LOTUS)	$E[g(X)] = \sum_x g(x) P(X = x)$ or $E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx$
Expectation of a Constant	$E[c] = c$
Linearity of Expectation	$E[aX + bY] = aE[X] + bE[Y]$
Expectation of Sums	$E\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} E[X_i]$
Expectation of Products (Independent Variables)	$E[XY] = E[X]E[Y]$
Monotonicity	$X \leq Y \Longrightarrow E[X] \leq E[Y]$
Non-negativity	$X \geq 0 \Longrightarrow E[X] \geq 0$
Absolute Value Inequality	$\|E[X]\| \leq E[\|X\|]$
Indicator Variables	$E[I_A] = P(A)$
Sample Mean is Unbiased	$E[\bar{X}] = \mu$

These properties constitute the algebraic foundation of the expectation operator and are among the most frequently used results in probability, statistics, stochastic processes, machine learning, econometrics, and statistical inference.

Most advanced results involving expectations—including variance, covariance, moment generating functions, laws of large numbers, the Central Limit Theorem, and estimation theory—are direct consequences or extensions of these fundamental properties.